PERSONAL FINANCIAL BUDGET ANALYSIS REPORT

Data Science Portfolio Project

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Tools Used: Microsoft Excel | Pivot Tables | Dashboard | Slicers

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Dataset: Personal Financial Transactions

Records Analyzed: 806 Transactions

Period Covered: January 2018 – September 2019

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2.EXECUTIVE SUMMARY

This report presents a comprehensive analysis of personal financial transactions covering the period January 2018 to September 2019. The dataset contains **806 transactions** across **19 spending categories** over a period of **21 months**, with a total expenditure of **$62,964.67** against a planned budget of **$49,371.00**, resulting in a net budget variance of **-$13,593.67**.

The analysis reveals that the majority of spending categories **13 out of 19** remained within their allocated budgets, demonstrating generally disciplined financial management. However, **6 categories exceeded their budgets**, with Home Improvement representing the most critical overspend at **$13,842.87 above budget**, which alone accounts for virtually the entire overall budget deficit.

Mortgage & Rent recorded the second highest overspend at **$1,654.50 above budget**, followed by Mobile Phone at **$315.40**, Gas & Fuel at **$140.17**, Electronics & Software at **$719.00** and Fast Food at **$15.63**. Notably, Electronics & Software and Movies & DVDs had no allocated budget yet recorded actual spending, indicating unplanned expenditure in these categories.

On the positive side, Entertainment, Alcohol & Bars and Restaurants all recorded significantly lower spending than budgeted, suggesting conservative lifestyle choices in discretionary categories. The overall net balance of **$28,185.98** between total income of **$124,269.76** and total expenses of **$96,083.78** confirms that despite the budget overruns, the overall financial position remains healthy and positive.

Based on these findings this report provides actionable recommendations to help improve budget adherence, manage the largest overspending categories and strengthen overall financial planning.

3. DATA OVERVIEW

3.1 Dataset Description

This analysis is based on two datasets extracted from a personal financial management system. The first dataset contains detailed transaction records covering all debit and credit activities across multiple accounts over a 21month period from January 2018 to September 2019. The second dataset contains monthly budget allocations across 19 spending categories used as the benchmark for performance evaluation.

3.2 Dataset Structure

Dataset 1 Personal Transactions

The transactions dataset contains **806 rows** and **6 original columns** which were expanded to **10 columns** after adding helper columns for deeper analysis:

Column	Type	Description
Date	Date	Date the transaction occurred
Description	Text	Merchant or transaction name
Amount	Number	Transaction amount in dollars
Transaction.Type	Text	Debit (money out) or Credit (money in)
Category	Text	Spending category assigned to transaction
Account.Name	Text	Account used for transaction
Month	Text	Month name extracted from Date
Year	Number	Year extracted from Date
Debit.Amount	Number	Amount if transaction is debit otherwise zero
Credit.Amount	Number	Amount if transaction is credit otherwise zero

Dataset 2 Budget Allocations

The budget dataset contains **19 rows** and **2 original columns** expanded to **7 columns** after analysis:

Column	Type	Description
Category	Text	Spending category name
Budget	Number	Monthly budget allocation per category
Total Budget	Number	Budget multiplied by 21 months
Actual Spent	Number	Total actual spending per category
Variance	Number	Total Budget minus Actual Spent
Status	Text	Under Budget or Over Budget
Variance %	Number	Variance as percentage of Total Budget

3.3 Data Preparation

The following preparation steps were performed before analysis:

- Date column was formatted from text to proper Excel date format
- Helper columns were added to separate debit and credit transactions
- Month and Year columns were extracted from the Date column using Excel TEXT and YEAR functions
- SUMIF function was used to calculate actual spending per category from 806 transactions
- Budget figures were multiplied by 21 months to create a fair comparison against actual spending
- Conditional formatting was applied to instantly highlight over and under budget categories

3.4 Key Dataset Statistics

Metric	Value
Total Transactions	806
Analysis Period	January 2018 – September 2019
Total Months Covered	21 months
Total Income	$124,269.76
Total Expenses	$96,083.78
Net Balance	$28,185.98
Total Budget Allocated	$49,371.00
Total Actual Spent	$62,964.67
Overall Budget Variance	-$13,593.67
Categories Under Budget	13 out of 19
Categories Over Budget	6 out of 19
Average Transaction Amount	$273.39

4. KEY FINDINGS

4.1 Overall Budget Performance

The overall budget analysis reveals that total actual spending of **$62,964.67** exceeded the total allocated budget of **$49,371.00** by **$13,593.67**. However this overrun is almost entirely driven by a single category Home Improvement which alone accounts for **$13,842.87** of overspending. Without Home Improvement the remaining 18 categories collectively performed within budget demonstrating strong overall financial discipline.

4.2 Category Performance

Out of 19 spending categories analyzed **13 categories remained under budget** and only **6 categories exceeded their budgets**. The 6 over budget categories are:

Category	Overspent By	Severity
Home Improvement	$13,842.87	Critical
Mortgage & Rent	$1,654.50	High
Electronics & Software	$719.00	Medium
Mobile Phone	$315.40	Medium
Gas & Fuel	$140.17	Medium
Fast Food	$15.63	Low

Notably Electronics & Software and Movies & DVDs had **zero budget allocation** yet recorded actual spending indicating completely unplanned expenditure in these categories.

4.3 Strong Performing Categories

Several categories demonstrated excellent budget management with significant underspending. Entertainment spent only **$9.62** against a **$525** budget 98% under. Alcohol & Bars spent **$539.13** against a **$1,050** budget saving **$510.87**. Restaurants saved **$536.98** below budget and Groceries saved **$354.79** below budget. These results reflect disciplined spending habits in discretionary categories.

4.4 Monthly Spending Pattern

Monthly spending analysis reveals clear fluctuations throughout the 21 month period. Certain months recorded significantly higher spending than others corresponding to periods of major home improvement activity and large one-time payments. These seasonal patterns highlight the importance of maintaining a financial buffer for unexpected high-spending months.

4.5 Income vs Expenses

Despite the budget overruns the overall financial position remains healthy. Total income of **$124,269.76** significantly exceeded total expenses of **$96,083.78** resulting in a positive net balance of **$28,185.98**. This confirms that while certain categories exceeded their budgets the overall financial situation is stable and manageable.

5. BUSINESS RECOMMENDATIONS

Based on the findings presented in this report the following recommendations are proposed to improve budget adherence and strengthen overall financial management:

5.1 Investigate and Control Home Improvement Spending

Finding: Home Improvement exceeded its budget by **$13,842.87** representing 263% over the allocated budget and accounting for virtually the entire overall budget deficit.

Recommendation:

- Investigate the specific transactions that drove this overspend was it a planned renovation or an emergency repair?
- If renovations are planned for the future create a **separate dedicated savings fund** specifically for home improvement projects
- Set a realistic monthly budget for Home Improvement based on historical spending — the current $250 monthly allocation is clearly insufficient
- For any home improvement project above **$1,000** require a minimum of **3 contractor quotes** before committing

Expected Impact: Properly budgeting for home improvement alone would eliminate the entire budget deficit and bring overall finances back on track.

5.2 Review Mortgage & Rent Commitment

Finding: Mortgage & Rent exceeded budget by **$1,654.50** over 21 months approximately **$79 per month** over the allocated amount.

Recommendation:

- Review the mortgage or rental agreement to understand if a rate increase occurred during the period
- Adjust the monthly budget allocation from **$1,100 to at least $1,180** to reflect actual payment amounts
- Explore mortgage refinancing options if interest rates have decreased since the original agreement was signed
- Consider making additional principal payments during high income months to reduce long term mortgage costs

Expected Impact: Adjusting the budget to reflect real mortgage costs will eliminate this recurring variance and improve budget accuracy.

5.3 Create Budgets for Unplanned Categories

Finding: Electronics & Software spent **$719.00** and Movies & DVDs spent **$222.19** with both having **zero budget allocation** meaning these were completely unplanned expenses.

Recommendation:

- Allocate a realistic monthly budget for Electronics & Software of at least **$35 per month** based on actual spending
- Allocate a monthly budget for Movies & DVDs of at least **$11 per month**
- Create a general **"Miscellaneous"** budget category of $50 per month to capture any future unplanned spending

- Review all 19 budget categories annually and add new categories as spending patterns evolve

Expected Impact: Having budgets for all active spending categories will give a more accurate picture of financial performance and eliminate surprise deficits.

5.4 Optimize Mobile Phone and Gas Expenses

Finding: Mobile Phone exceeded budget by **$315.40** and Gas & Fuel exceeded budget by **$140.17** over the 21 month period.

Recommendation:

- Review current mobile phone plan and compare with alternative providers significant savings are often available by switching plans or providers
- Consider carpooling, combining errands or using fuel rewards programs to reduce Gas & Fuel costs
- If driving patterns have permanently increased adjust the Gas & Fuel budget from **$75 to at least $82 per month** to reflect actual usage
- Track fuel purchases weekly to identify any unusual spikes in consumption

Expected Impact: Optimizing these two categories could recover up to **$455 annually** in unnecessary spending.

5.5 Maintain Strong Performance in Well Managed Categories

Finding: 13 categories remained under budget with Entertainment, Alcohol & Bars and Restaurants showing particularly strong discipline.

Recommendation:

- Continue the current disciplined approach to discretionary spending in categories like Entertainment, Restaurants and Groceries
- Consider redirecting the savings from under budget categories into a **dedicated emergency fund**

- Set a savings target of at least **$500 per month** using the surplus from well managed categories
- Review under budget categories annually to ensure allocations remain realistic and not unnecessarily restrictive

Expected Impact: Redirecting budget surpluses from well managed categories into savings could generate **$6,000+ in annual savings.**

5.6 Build a Monthly Financial Review Habit

Finding: Several budget overruns particularly in Home Improvement could have been detected and managed earlier with regular monitoring.

Recommendation:

- Schedule a **monthly 30minute financial review** to compare actual spending against budget
- Use the Excel dashboard built in this project to monitor performance in real time
- Set up **spending alerts** on all bank accounts and credit cards for transactions above $500
- Create a simple traffic light system Green for under budget, Yellow for within 10% of budget, Red for over budget and review it every month

Expected Impact: Monthly monitoring catches problems early before they become large deficits potentially saving thousands of dollars annually.

6. CONCLUSION

This report has presented a comprehensive analysis of personal financial transactions covering 21 months from January 2018 to September 2019. Through systematic examination of 806 transactions across 19 spending categories this analysis has revealed both the strengths and weaknesses of the current financial management approach.

The overall financial position is fundamentally healthy. Total income of **$124,269.76** significantly exceeded total expenses of **$96,083.78** generating a positive net balance of **$28,185.98** over the analysis period. This confirms that the individual is living within their overall means despite certain category level budget overruns.

The analysis identified that **13 out of 19 categories** performed within their allocated budgets demonstrating strong financial discipline in the majority of spending areas. Discretionary categories such as Entertainment, Restaurants and Alcohol & Bars showed particularly impressive budget management with significant underspending recorded across all three.

However the analysis also uncovered critical areas requiring immediate attention. The Home Improvement category alone overspent by **$13,842.87** accounting for virtually the entire overall budget deficit of **$13,593.67**. This single category represents the most urgent financial planning challenge identified in this report. Additionally Mortgage & Rent, Mobile Phone and Gas & Fuel all recorded consistent monthly overruns that collectively add up to meaningful amounts over time.

Perhaps most importantly the analysis revealed that two categories Electronics & Software and Movies & DVDs had **no budget allocation** despite recording actual spending. This highlights a gap in the budgeting process that must be addressed to ensure all spending is properly planned and tracked going forward.

By implementing the six recommendations outlined in this report investigating Home Improvement spending, reviewing the mortgage commitment, creating budgets for unplanned categories, optimizing Mobile Phone and Gas expenses, maintaining strong performance in well managed categories and building a monthly financial review habit this individual is well positioned to achieve full budget adherence and accelerate their financial goals in the years ahead.

The interactive Excel dashboard developed as part of this project provides a powerful ongoing tool for monthly financial monitoring and will continue to deliver value well beyond the scope of this analysis.